DIPLOMA PROGRAMME PRACTICE · HL · SET A

Biology / Paper 1A + 1B

Worked mark scheme · 75 marks · 120 minutes across 1A and 1B

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each.

PAPER 1A · QUESTION 1 [1]

Answer C Temperature and pressure determine liquid stability, while adequate gravity helps retain atmospheric and water molecules.

PAPER 1A · QUESTION 2 [1]

Answer D The other 126 pairs are A–T, each containing exactly one thymine.

PAPER 1A · QUESTION 3 [1]

Answer B Polymerase extends the growing strand at its 3′ end, forming a phosphodiester bond in its backbone.

PAPER 1A · QUESTION 4 [1]

Answer B Self-assembly supplies a plausible compartment. It does not demonstrate all the functions of the first cells.

PAPER 1A · QUESTION 5 [1]

Answer A Integration and copying with the host characterize lysogeny. Assembly and lysis identify productive lytic infection.

PAPER 1A · QUESTION 6 [1]

Answer D The biological species concept relies on reproductive isolation. A difference in appearance, geography or allele frequency alone does not establish this.

PAPER 1A · QUESTION 7 [1]

Answer A Shared derived states support an X–Y clade. Independent changes or reversals could weaken the inference.

PAPER 1A · QUESTION 8 [1]

Answer C A ploidy difference can disrupt pairing in crosses with the original population while permitting reproduction among compatible polyploids.

PAPER 1A · QUESTION 9 [1]

Answer B Sampling unrelated individuals across the range reduces the loss of alleles through founder effects.

PAPER 1A · QUESTION 10 [1]

Answer C Activity depends on shape. Denaturation can disrupt interactions maintaining that shape while leaving the primary sequence intact.

PAPER 1A · QUESTION 11 [1]

Answer D A charged group in a hydrophobic environment can alter energetically favourable folding, with effects on distant residues.

PAPER 1A · QUESTION 12 [1]

Answer C Cholesterol disrupts ordered packing in the cold and can restrain phospholipid movement at higher temperatures.

PAPER 1A · QUESTION 13 [1]

Answer D A secreted protein enters rough ER, is processed and sorted in Golgi, and reaches the membrane in a vesicle.

PAPER 1A · QUESTION 14 [1]

Answer C Inner membrane area sets potential capacity. Substrate supply, oxygen and demand also affect the actual rate.

PAPER 1A · QUESTION 15 [1]

Answer D At fixed area and concentration difference the rate varies inversely with thickness.

PAPER 1A · QUESTION 16 [1]

Answer C Acidification shifts the dissociation curve to favour oxygen unloading, the Bohr effect.

PAPER 1A · QUESTION 17 [1]

Answer D The ventricle has closed the inlet valve but has not produced enough pressure to open the outlet valve.

PAPER 1A · QUESTION 18 [1]

Answer D Sliding increases overlap. A band length reflects thick filament length and remains approximately constant.

PAPER 1A · QUESTION 19 [1]

Answer A Still, humid air near the stomata reduces the gradient that drives evaporation from the leaf.

PAPER 1A · QUESTION 20 [1]

Answer D Subtract non-enzymatic loss from the 1.8 mmol total and divide 1.5 mmol by 30 s.

PAPER 1A · QUESTION 21 [1]

Answer A High substrate concentration can outcompete a reversible competitive inhibitor, leaving the attainable maximum unchanged.

PAPER 1A · QUESTION 22 [1]

Answer C Respiration transfers some substrate energy to ATP and some to the surroundings as heat.

PAPER 1A · QUESTION 23 [1]

Answer B A leak dissipates the electrochemical gradient without driving ATP synthase, lowering coupling efficiency.

PAPER 1A · QUESTION 24 [1]

Answer D Light-driven electron flow reduces $NADP^+$ and supports ATP synthesis through a proton gradient.

PAPER 1A · QUESTION 25 [1]

Answer A A surface receptor can relay an extracellular binding event through second messengers or protein interactions.

PAPER 1A · QUESTION 26 [1]

Answer B An action potential propagates as successive regions of the membrane change electrical potential.

PAPER 1A · QUESTION 27 [1]

Answer A Depolarization at a node normally triggers Na^+ influx and regenerates a full action potential.

PAPER 1A · QUESTION 28 [1]

Answer A Differential auxin distribution can cause differential elongation in shoots and curvature toward light.

PAPER 1A · QUESTION 29 [1]

Answer B Marked proportion in the sample is 12/60. Equating that to 80/N gives N=400.

PAPER 1A · QUESTION 30 [1]

Answer C After two rounds there are four molecules per original, of which two retain one original labelled strand.

PAPER 1A · QUESTION 31 [1]

Answer B Starting copies are multiplied by 2^6=64, giving 15×64=960.

PAPER 1A · QUESTION 32 [1]

Answer A Splice sites guide removal of introns from pre-mRNA. Comparing mature transcripts tests the proposed processing defect.

PAPER 1A · QUESTION 33 [1]

Answer C Comparing sequence changes at other potential targets with matched controls tests unintended cutting.

PAPER 1A · QUESTION 34 [1]

Answer B Replication produces two sister chromatids per chromosome. Chromosome number is counted by centromeres before separation.

PAPER 1A · QUESTION 35 [1]

Answer B Transcription factors can alter initiation frequency without changing the coding sequence.

PAPER 1A · QUESTION 36 [1]

Answer D Cell water potential is −0.50 MPa. Water moves from higher (less negative) potential to lower potential.

PAPER 1A · QUESTION 37 [1]

Answer C Each offspring has probability ½ of inheriting the recessive allele from the heterozygote. Independent events give ½×½.

PAPER 1A · QUESTION 38 [1]

Answer B The nonparental types total 200 of 1000 offspring. Recombination frequency is 20%.

PAPER 1A · QUESTION 39 [1]

Answer A ADH increases water reabsorption, allowing more concentrated urine and opposing the rise in plasma osmolarity.

PAPER 1A · QUESTION 40 [1]

Answer A $q^2=0.09$ gives $q=0.30$ and $p=0.70$. The heterozygote frequency is $2pq=0.42$.

PAPER 1B · QUESTION 1 [8]

Interpreting an electron micrograph

(a) Identify the large organelle and give one visible feature supporting the identification. [2]

1 It is a mitochondrion.

1 Internal folds consistent with cristae are visible inside the surrounding envelope.

(b) Estimate its maximum visible length relative to the scale bar and give the approximate length. [2]

1 The long dimension is roughly 2.5 scale bars (accept 2.3–2.8).

1 Each bar is 1.0 µm, so the length in section is about 2.5 µm with consistent measurement.

(c) Compare the two normalized intersection counts. Why does the larger raw count alone not establish a difference in membrane density? [2]

1 $X=48/40=1.20$ and $Y=90/60=1.50$ intersections μm^{-1}.

1 Y is 25% higher after normalization, but more total line length was sampled in Y.

(d) Explain why the count alone does not establish a higher ATP synthesis rate in the intact tissue. [2]

1 The count provides a structural estimate whose interpretation depends on sampling and section orientation.

1 Actual synthesis also depends on factors such as oxygen availability, ADP supply and respiratory enzyme activity.

PAPER 1B · QUESTION 2 [9]

Oxygen delivery under two conditions

(a) At 4 kPa calculate O_2 unloaded per dm^3 blood at each tissue pH. [3]

1 At 7.4 the saturation drop is 97−55=42 percentage points, giving 84 cm^3 dm^{-3}.

1 At 7.2 the drop is 97−40=57 points, giving 114 cm^3 dm^{-3}.

1 The lower pH permits 30 cm^3 more O_2 to unload per dm^3.

(b) Relate the change to respiration in an active tissue. [3]

1 Respiration produces CO_2, contributing to increased H^+ and lower pH.

1 Lower pH reduces haemoglobin affinity for oxygen at a given partial pressure.

1 More oxygen is released where tissue oxygen demand is high.

(c) Blood flow in a muscle is 0.60 dm^3 min^{-1} at the lower pH. Estimate delivery to the tissue per minute and give two assumptions behind applying the calculation. [3]

1 0.60×114=68.4 cm^3 O_2 min^{-1}.

1 Assume outgoing blood equilibrates with the specified tissue oxygen pressure and pH.

1 Assume the given inlet saturation and capacity are appropriate and flow is steady; accept another stated modelling assumption.

PAPER 1B · QUESTION 3 [9]

Correcting a respirometer

(a) Calculate corrected oxygen uptake per gram per minute over 12 min. [3]

1 Corrected displacement=18−3=15 mm.

1 Volume=15×0.40=6.0 mm^3.

1 Rate=6.0/(1.60×12)=0.313 mm^3 g^{-1} min^{-1}.

(b) Explain the role of the absorbent and the bead apparatus. [3]

1 Respiration consumes O_2 and produces CO_2.

1 Removing CO_2 allows corrected gas-volume loss to represent oxygen uptake.

1 Beads reveal displacement caused by external pressure or temperature changes in the apparatus.

(c) A second set at 30 °C has a higher corrected rate. State a suitable experimental improvement and explain why this result does not prove a higher ATP yield per oxygen molecule. [3]

1 Use independent seed batches at each temperature and compare variability while holding mass and germination stage constant.

1 Uptake measures oxygen consumption rate, not directly ATP production.

1 ATP yield per O_2 also depends on coupling, so ATP production would need an independent measurement.

PAPER 1B · QUESTION 4 [9]

A test cross with unequal offspring counts

(a) Identify the likely parental arrangement and estimate the recombination frequency. [3]

1 MN and mn are the most frequent, supporting MN/mn in the heterozygote.

1 Recombinant offspring=66+54=120 of 500.

1 Observed recombination=24%.

(b) What counts would independent assortment predict, assuming equal survival? Use the observed pattern to assess that model. [3]

1 The expected ratio is 1:1:1:1, or 125 in each class.

1 Both parental classes exceed expectation while both recombinant classes fall below it.

1 The pattern supports linkage, although statistical testing and examination of survival bias strengthen the inference.

(c) Explain how crossing over makes the less common classes and give one reason the observed percentage need not equal the exact fraction of meiotic cells with any crossover. [3]

1 Non-sister chromatids of homologues exchange corresponding segments during prophase I.

1 An exchange between the two loci can produce Mn and mN combinations.

1 A crossover involves only two of four chromatids, and multiple crossovers can also restore parental marker combinations.